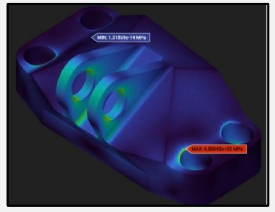
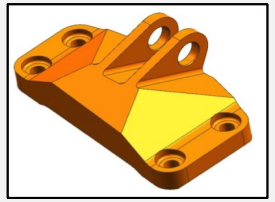
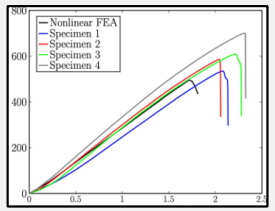


$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

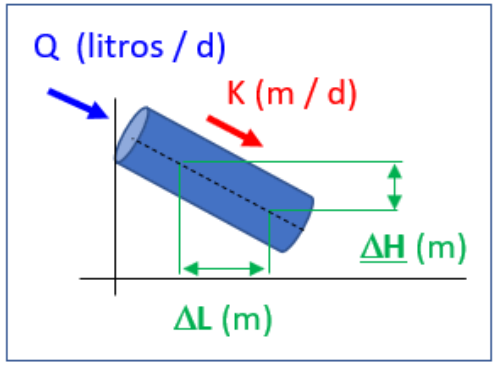






Experimento de **Darcy:**

Base de hidrogeologia

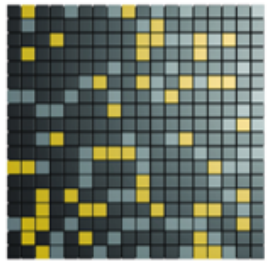


Velocidade de Darcy:

q (m/d)

Vazão / área (seção ∅) = Condutividade x Gradiente

$$\frac{Q}{A_{\varnothing}} = q = k\frac{\Delta H}{\Delta L}$$



2D



3D

$$\dot{Q} = \frac{\kappa A \Delta T}{\ell}$$

Calor

Condução

Fourier

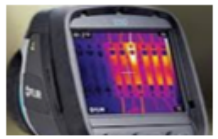
Convecção

Radiação

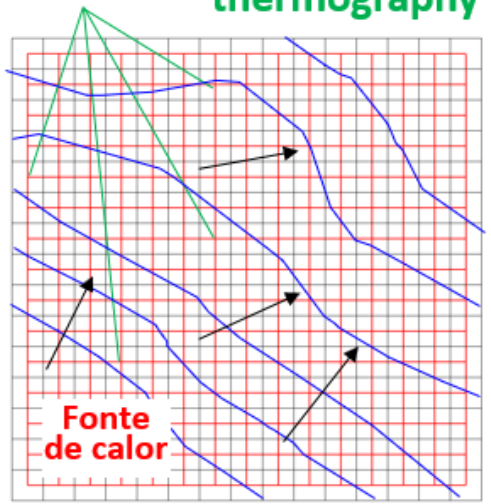
Mosaico (grid) de pequenas placas de metal de variada composição



Termopar



infrared thermography



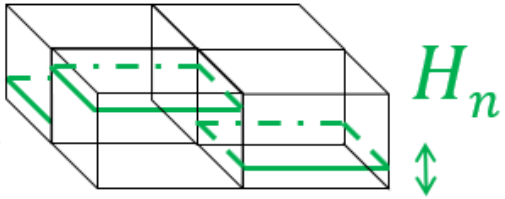
Modelo numérico

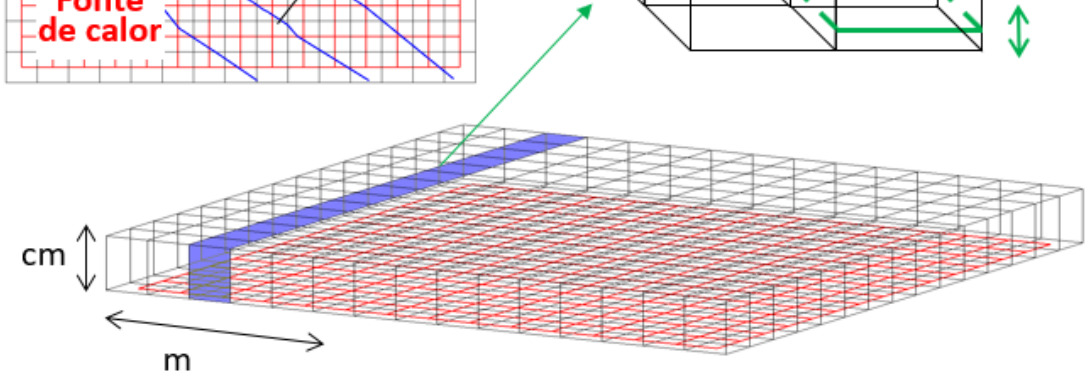
VS

Experimento Físico

(Mosaico impressão 4D)

Temperatura = carga hidráulica





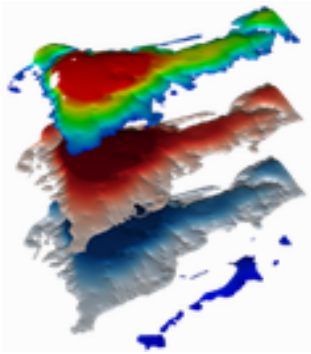
Resultado do experimento físico: Exemplo de base de dados para calibração do modelo

Aplicação da metodologia MODFLOW|PEST ao problema inverso na calibração de projetos da Fluidodinâmica Computacional

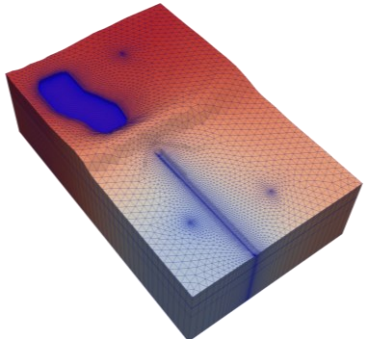
Formulação sintética: Transferência de Calor

USGS
U.S. Geological Survey

FloPy$_3$

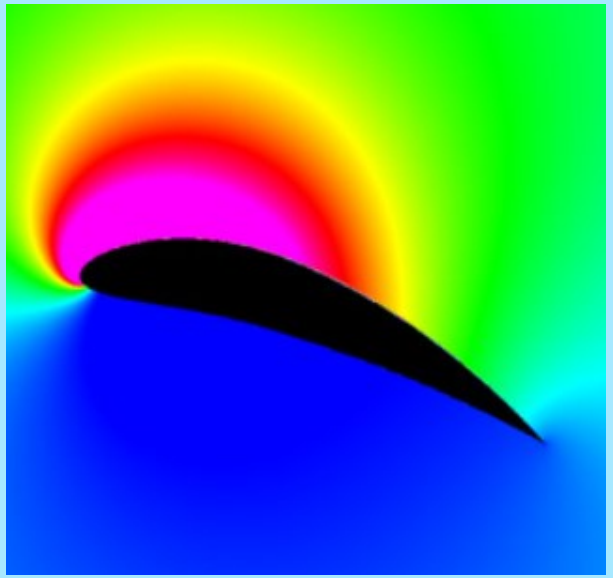


MODFLOW 6



PEST
Model-Independent
Parameter
ESTimation

CFD **MA**

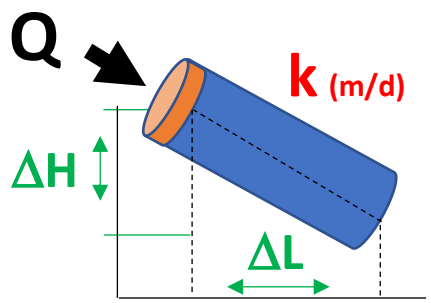


Candidato:
Eng. MSc
Rodrigo M. Grossi

Orientador:
Prof. Doutor
Andreas Nascimento

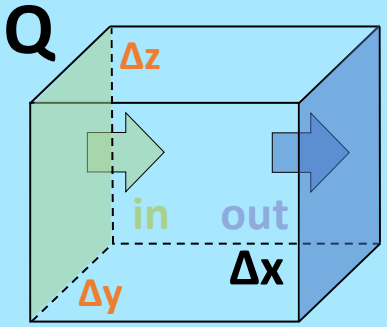
Contexto

Darcy ~ Fourier



$$\frac{Q}{A_{\varnothing}} = q = k\frac{\Delta H}{\Delta L}$$ H_2O

$$\dot{Q} = (-kA\,\Delta T)/L$$ Calor



$$\frac{\partial q_x}{\partial x}\,\Delta x\,\Delta y\Delta z \quad \ldots \quad \Delta x\ \Delta y\ \Delta z \quad \Delta x\ \Delta y\ \Delta z$$

infinitesimal

$$\frac{\partial^2 h}{\partial x^2} + \frac{\partial^2 h}{\partial y^2} + \frac{\partial^2 h}{\partial z^2} = Q + PA$$

Laplace

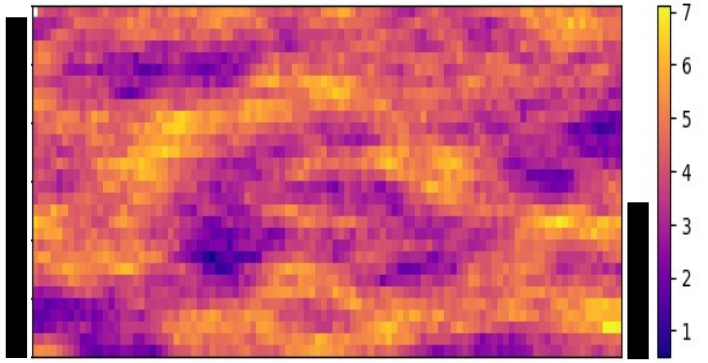
$$\nabla^2 h$$ nabla

= fontes
(-) sumidouros

$$\frac{\partial}{\partial x}\left(k\frac{\partial h}{\partial x}\right)$$

MODFLOW

Premissas



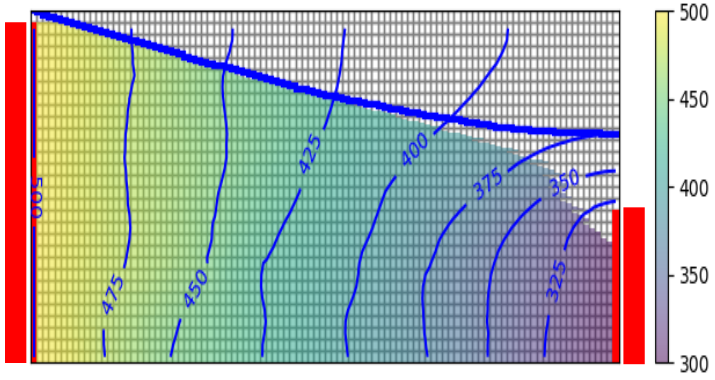
k BC

Condutividade

... Resistência

... Topologia (!)

Resultados



h

Carga (hidráulica)

Cálculo Bayesiano

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Inversão

“ Loop ” de retroalimentação

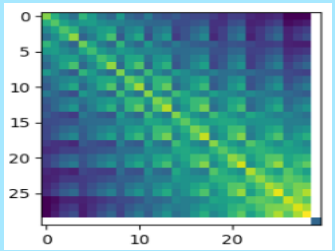
$$\frac{\partial \phi}{\partial t} + \nabla.(u\phi) - D\nabla^2\phi = S_\phi$$

convecção difusão fontes

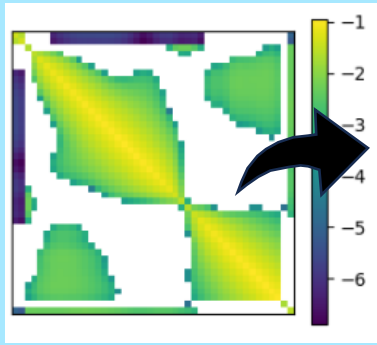
$$\begin{bmatrix} * & * & & * \\ * & * & * & \\ & * & * & * \\ * & & * & * \end{bmatrix} * \begin{bmatrix} \phi \end{bmatrix} = \begin{bmatrix} * \\ * \\ * \\ * \end{bmatrix}$$

$$A * \phi = b$$

Modelo OBS ⟷ PAR



COV



0.4	0.1	0.2
0.1	0.4	0.2
0.2	0.2	0.4

Zoom +

Condutividade hidráulica

x load flow

- Densidade (Massa)
- Elasticidade
- Rigidez
- Limite de escoamento

New!

Probabilidade condicional

$$\phi = A^{-1} b$$

Inversão

Cálculo Bayesiano

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

FOCO

Sparce matrix

Derivative matrix

Asymmetric matrix

COVariance matrix

Coefficient matrix

Regressor matrix

Design matrix

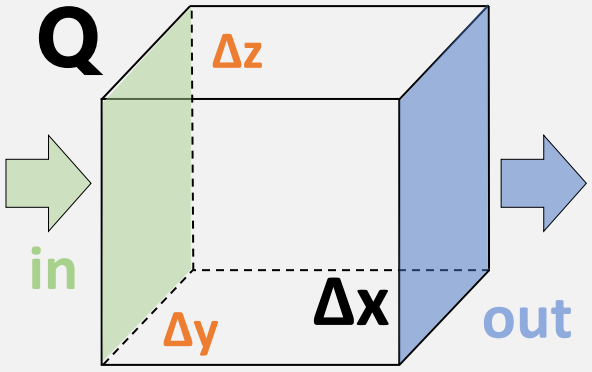
Jacobian matrix

Abordagem Estocástica

FEA – Análise de Elementos Finitos

- Decompõe um objeto 3D complexo em milhares de pequenas partes gerenciáveis ("elementos");

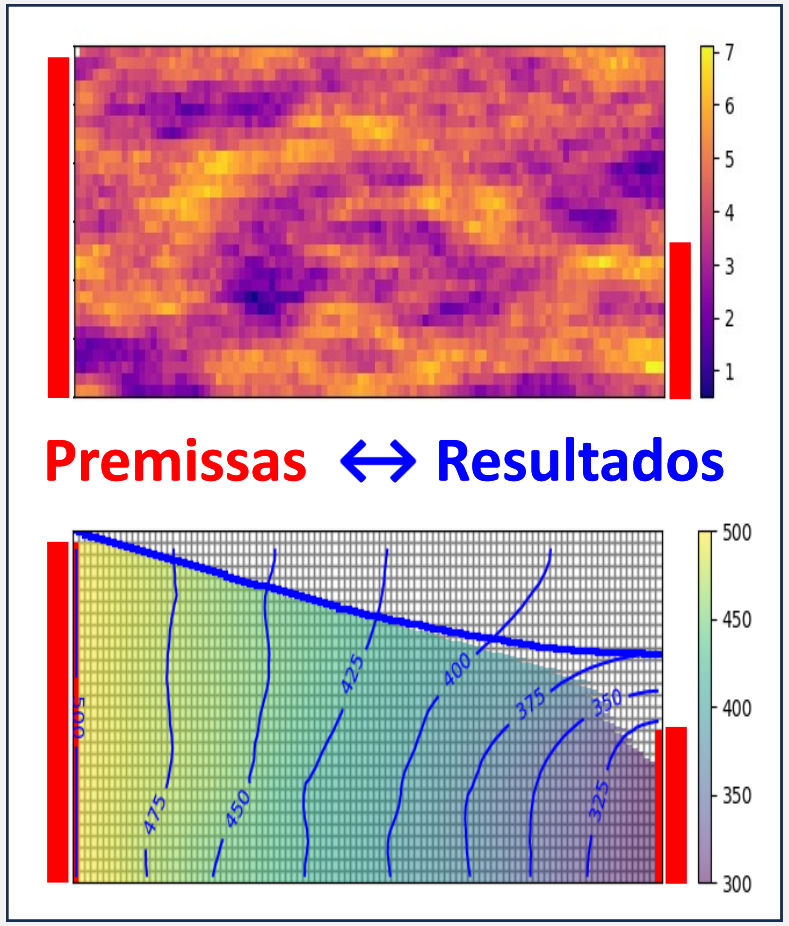
- Calcula tensões estruturais, falhas e deformações.

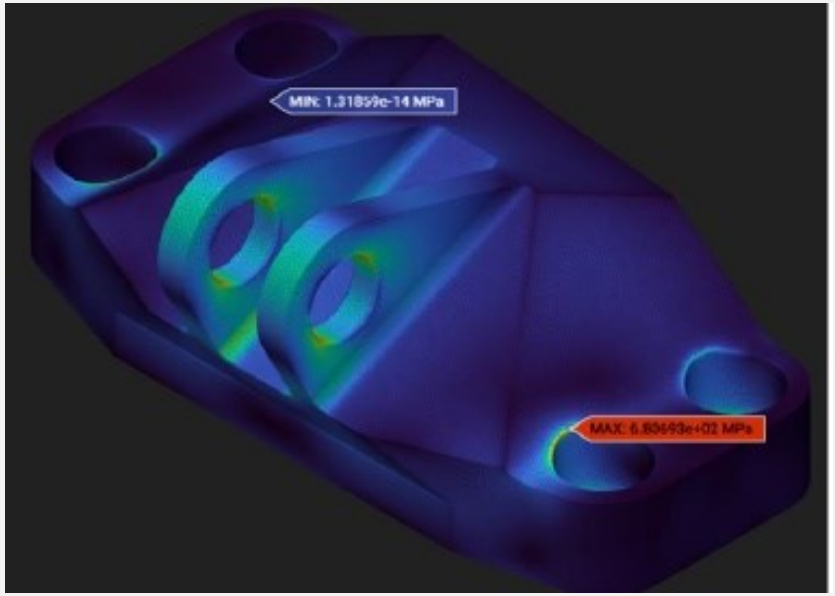


MODFLOW

$$\frac{\partial^2 h}{\partial x^2} + \frac{\partial^2 h}{\partial y^2} + \frac{\partial^2 h}{\partial z^2} = Q + PA$$

$$\nabla^2 h$$



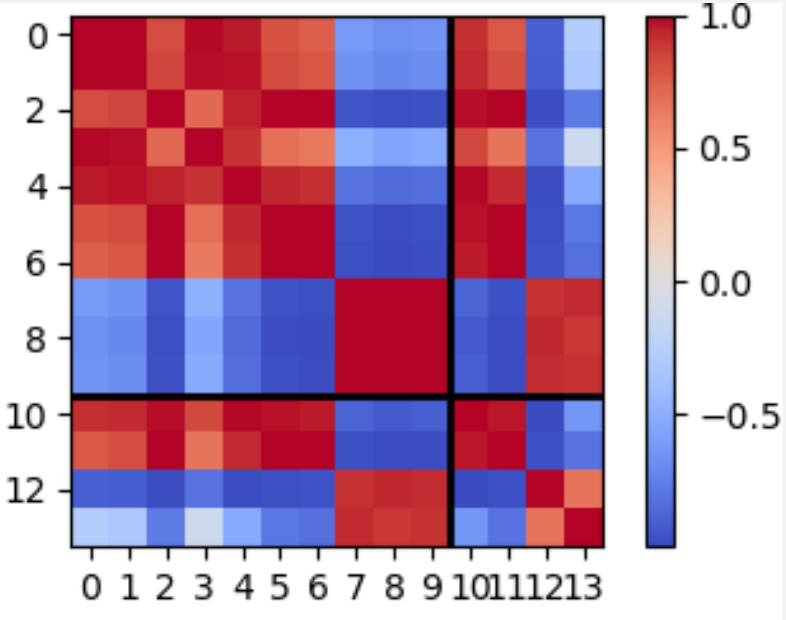


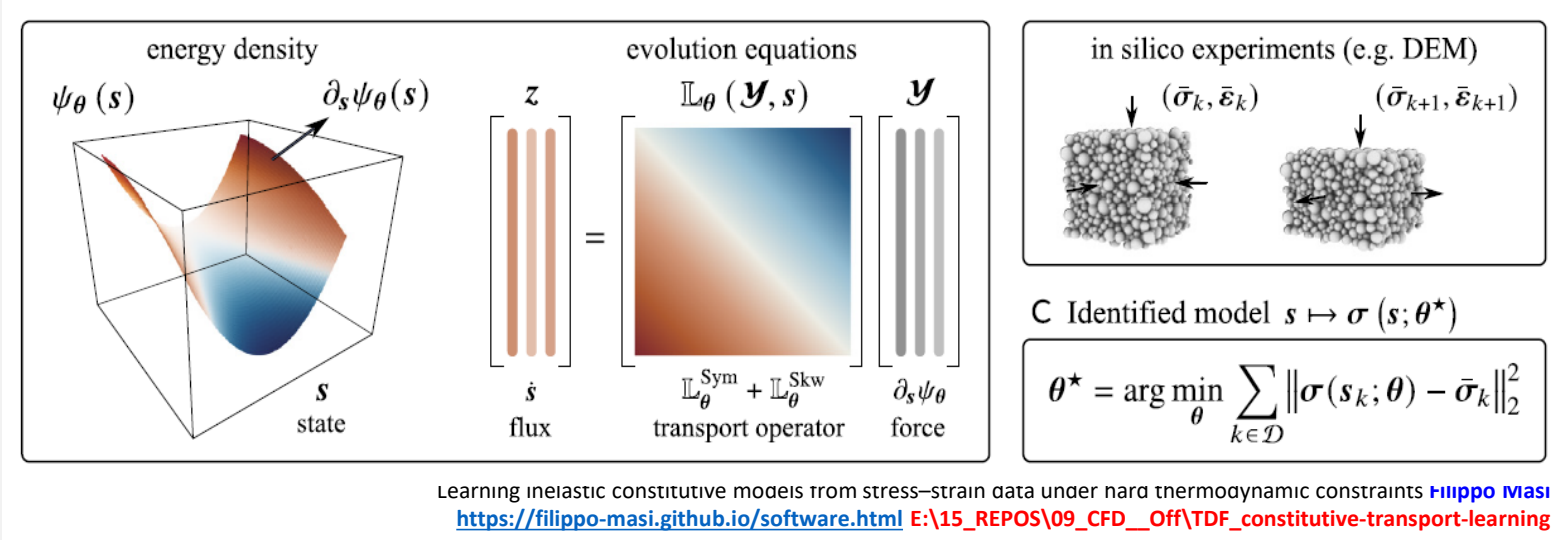


Peça de Suporte

Modelo = Matriz de covariâncias espaciais entre PAR vs OBS

XY





Learning inelastic constitutive models from stress–strain data under hard thermodynamic constraints Filippo Masi
https://filippo-masi.github.io/software.html E:\15_REPOS\09_CFD__Off\TDF_constitutive-transport-learning

$$X * B = \hat{Y}$$

Modelo IN | OBS ⟷ OUT



Peça de Suporte

EXEMPLO 05: *Functionally graded materials (FGMs)* - (FGAM)

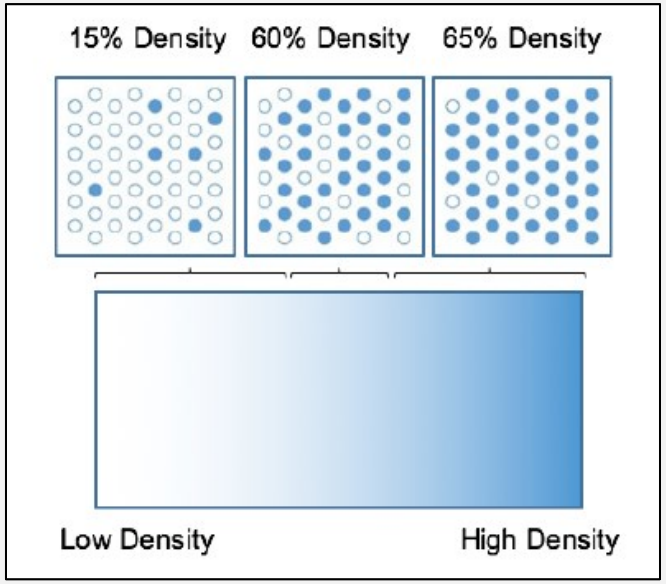
A.M. – strategically control

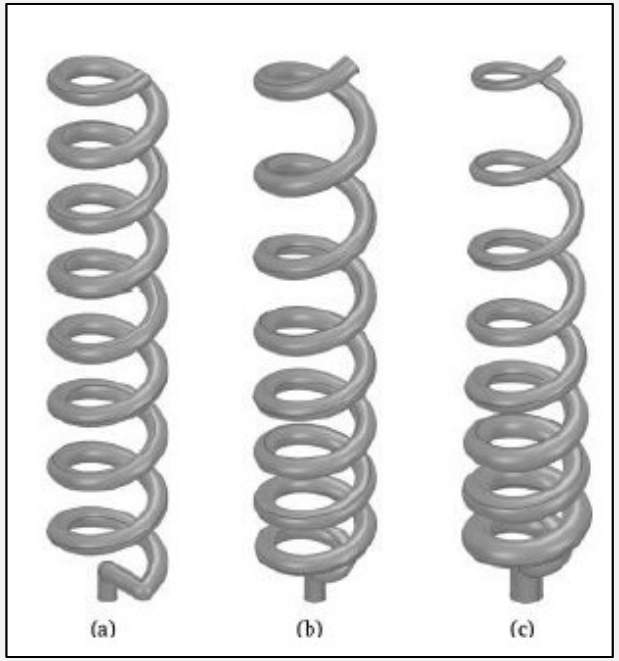
– density, directionality and allocation … of material substances

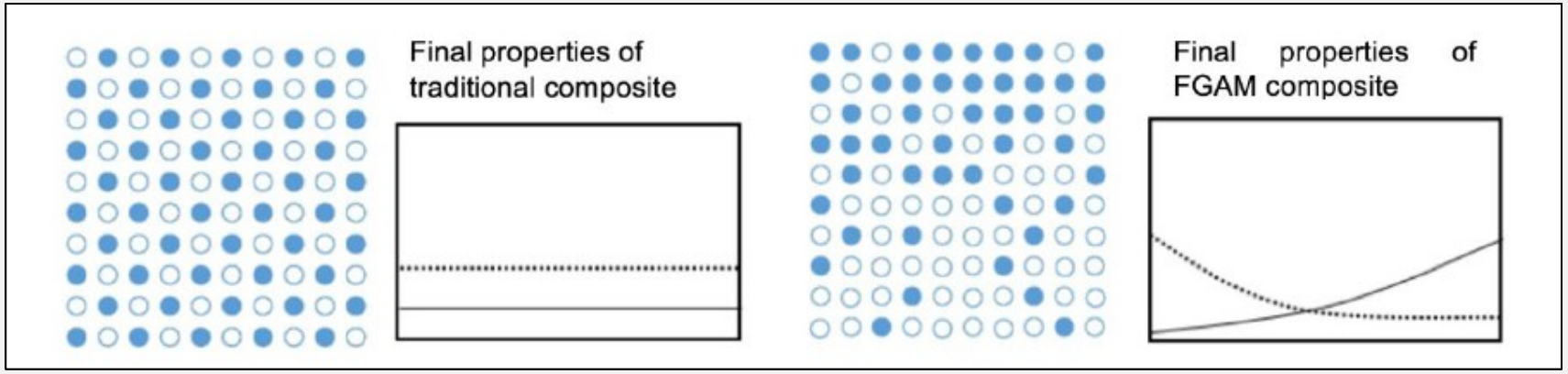
Commercial available AM Technologies …

…

… still operate predominantly on **isotropic materials**







EXEMPLO 05: *Functionally graded materials (FGMs)* - (FGAM)

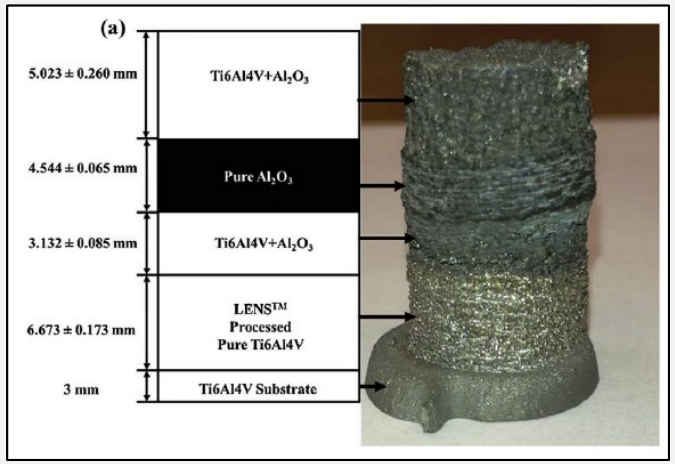
A.M. – strategically control
– density, directionality and allocation … of material substances

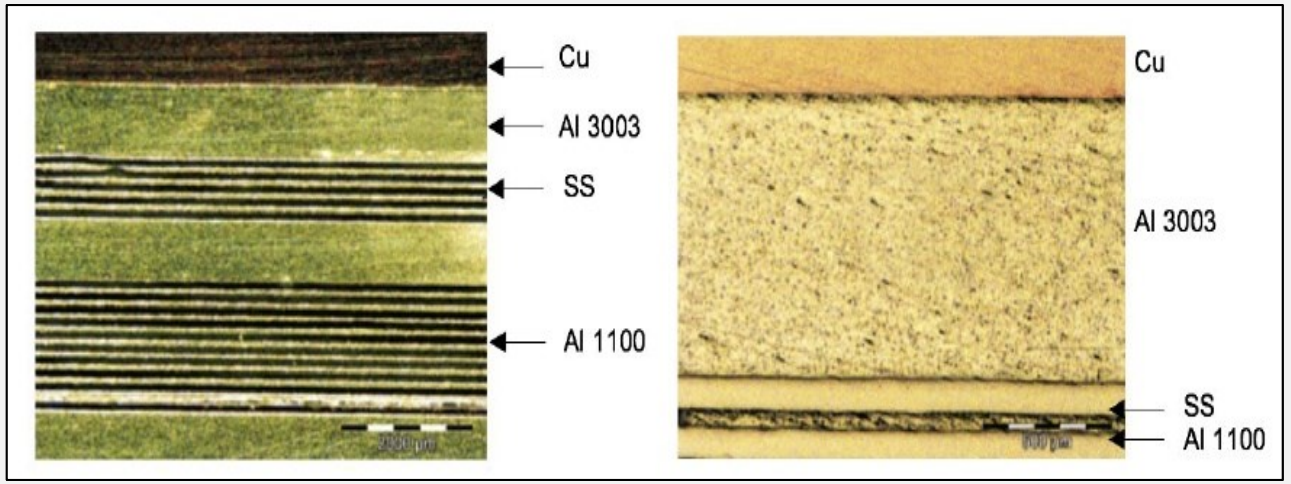
Complex FGMs with multi-dimensional and directional gradient structures



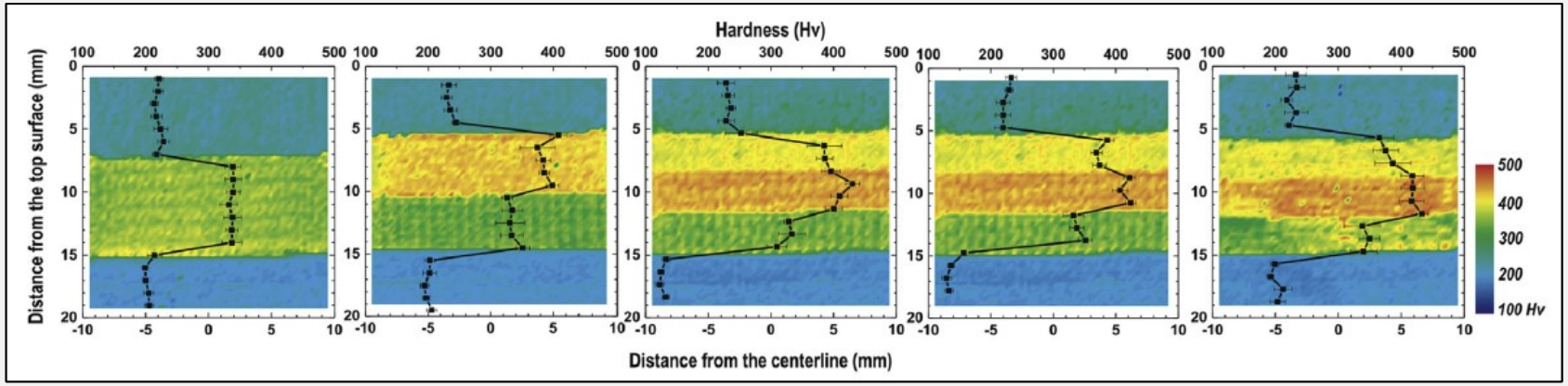


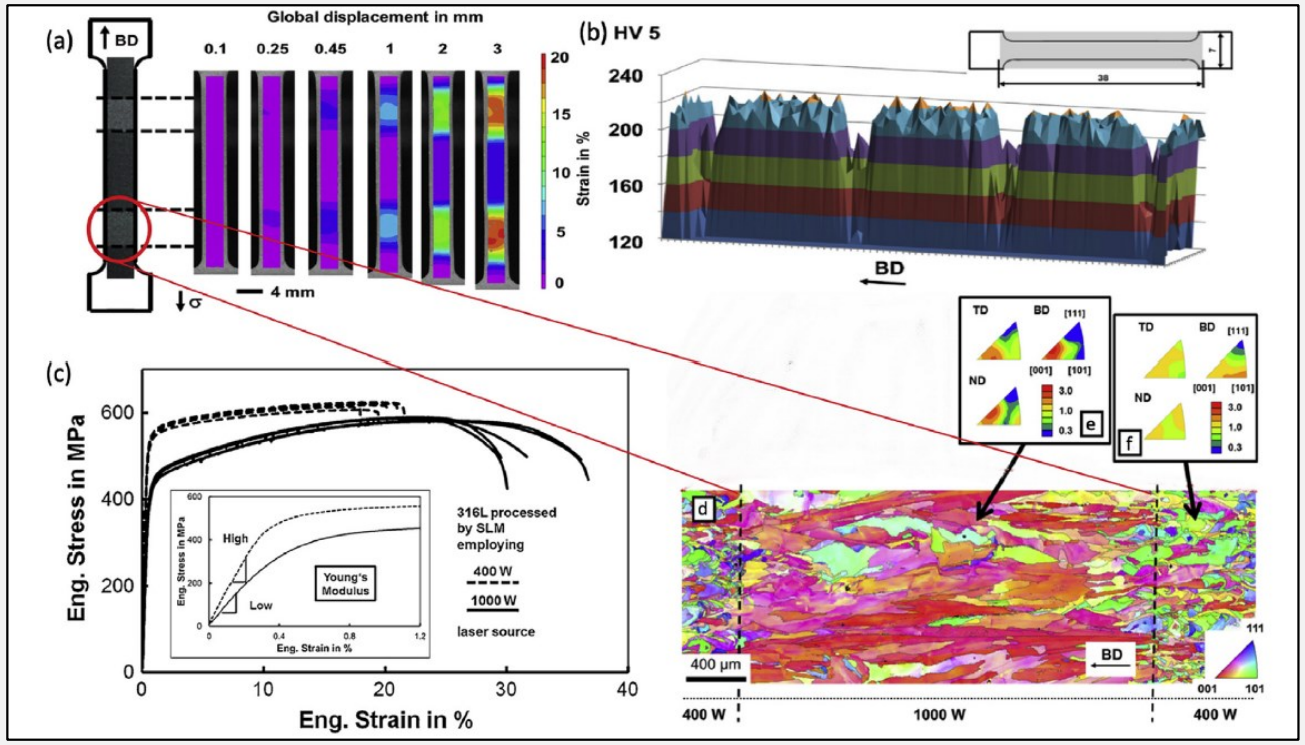
alloy

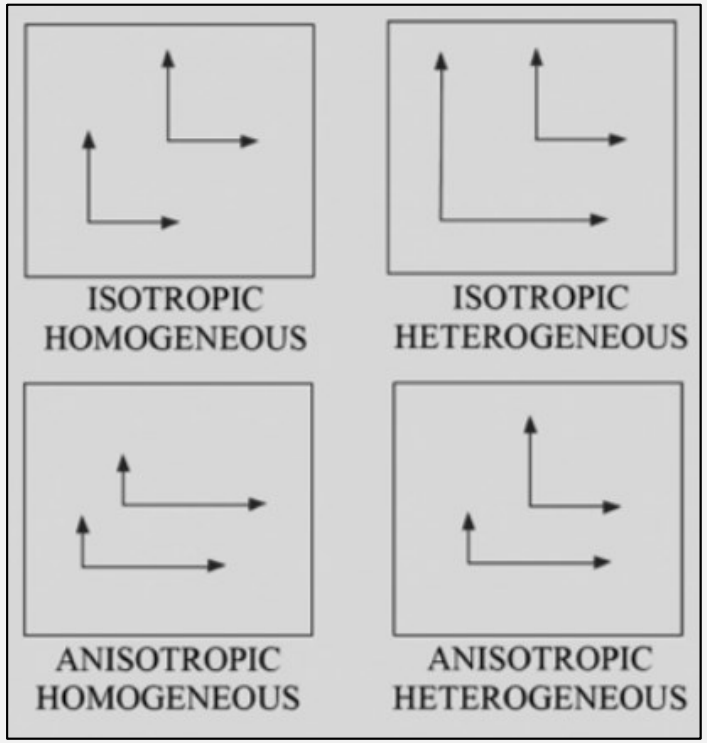




Heterogeneities
Anisotropy







Obrigado!

Rodrigo M. Grossi

https://grossirm.github.io/

https://betami2022.wixsite.com/betami